

A Modular ZK-Rollup Prototype for Benchmarking Layer-2 Scalability Bottlenecks

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Abstract—Ethereum’s limited Layer-1 throughput has motivated the adoption of Layer-2 rollups, where ZK-Rollups provide strong security and fast finality through validity proofs. However, practical ZK-Rollup performance depends heavily on configuration choices such as batch size, sequencing policy, data availability mode, and proof-generation overhead. Existing production rollups are difficult to modify for controlled experimentation, limiting reproducible analysis of these trade-offs.

This paper presents a modular ZK-Rollup prototype designed for benchmark-driven performance evaluation. The prototype separates the sequencer, executor, submitter, and prover components, enabling controlled experiments over different batching, scheduling, and data availability configurations. The prototype is evaluated using synthetic workloads, and key metrics such as throughput, latency, gas cost per transaction, batch finalization time, and prover-side overhead are analyzed.

The proposed modular evaluation framework isolates key bottlenecks in the rollup pipeline, exposing how configuration choices shift the throughput-latency-cost trade-off. By serving as a reproducible basis for studying ZK-Rollup scalability, the framework offers practical guidance for selecting optimal architectural parameters.

Index Terms—ZK-Rollups, Ethereum, Layer-2 Scaling, Blockchain Scalability, Benchmarking, Sequencer, Data Availability

I. INTRODUCTION

Ethereum Layer-1 provides strong security and decentralization but suffers from limited throughput, high congestion, and fee volatility [1], [2]. To address the blockchain trilemma, Layer-2 rollups have emerged as the primary scaling direction [3]. Among these, Zero-Knowledge (ZK) Rollups are particularly promising because they execute transactions off-chain and submit succinct cryptographic validity proofs to Layer-1, offering faster finality and stronger security guarantees than optimistic rollups [4].

However, practical ZK-Rollup performance depends heavily on interacting bottleneck factors, including proof-generation overhead, sequencer scheduling policies, data availability (DA) mode costs, and batch size/frequency trade-offs [5], [6]. Existing production ZK-Rollups are optimized for deployment and user adoption, but their monolithic complexity makes them difficult to modify for controlled academic benchmarking.

To address this gap, this paper presents a modular ZK-Rollup research prototype designed to separate sequencing, execution, proving, submission, and data availability concerns. By doing so, it supports reproducible, configuration-level

benchmarking. The system is designed as a controlled experimental environment and is not a production-ready system or a full zkEVM.

The main contributions of this work are:

- We design and implement a modular ZK-Rollup research prototype with separated sequencer, executor, prover, submitter, smart contract, and DA components.
- We expose configurable batching, sequencing, and DA policies to support controlled benchmarking.
- We define a reproducible methodology for evaluating throughput, latency, gas per transaction, proof or mock-proof time, finalization time, DA footprint, and queueing behavior.
- We analyze the resulting throughput-latency-cost trade-offs and identify bottlenecks that guide future ZK-Rollup optimization.

II. BACKGROUND AND RELATED WORK

A. Ethereum Scalability and Rollups

Ethereum’s rollup-centric roadmap relies on Layer-2 solutions to move execution off-chain while utilizing Layer-1 for settlement and data availability [7]. ZK-Rollups use validity proof systems (such as Groth16 or PLONK) to mathematically guarantee off-chain execution correctness [8], [9].

B. Performance Bottlenecks and Benchmarking

Recent benchmarking efforts highlight multiple bottlenecks in ZK-Rollups. Generating validity proofs remains a significant computational hurdle [5]. Furthermore, data availability poses a trade-off between cost and security; the introduction of EIP-4844 (proto-danksharding) aims to alleviate this by providing cheaper blob-based storage [10]–[12]. The design of the sequencer and the chosen transaction ordering policies also heavily influence latency and fairness [13]. Existing studies often analyze public datasets or production traces [6], [14].

C. Research Gap

While prior work examines rollup architectures, costs, data availability, sequencer decentralization, and production traces, there is still a need for a compact modular framework that supports controlled experiments over batching, sequencing, DA, and proof overhead under identical workloads.

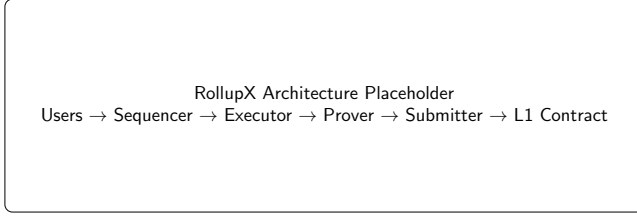


Fig. 1. Architecture of the proposed modular ZK-Rollup benchmarking prototype.

III. SYSTEM ARCHITECTURE

A. Overview

The proposed framework is designed as a modular ZK-Rollup research prototype. The architecture deliberately separates policy decisions from execution and settlement mechanisms. This design prioritizes observability and reproducibility over production completeness, avoiding the complexity of a full production zkEVM.

B. Transaction Lifecycle and Components

The end-to-end transaction lifecycle involves several decoupled components, as illustrated in Fig. 1.

Workload Generator: Produces synthetic transaction traffic at controlled rates, supporting various workload regimes such as balanced steady-state or bursty arrivals.

Sequencer: Maintains a transaction pool and orders transactions based on configurable scheduling policies (e.g., FCFS, fee-priority, time-boost, or fair BFT). It includes a batch orchestrator that determines when to form a new batch based on fixed-size, fixed-interval, or soft-cap rules. It also monitors a forced transaction queue to simulate censorship resistance.

Executor: The state manager applies state transitions to a sparse Merkle tree. It validates nonces and balances, separating execution from transaction ordering.

Prover: The mock-prover mode parameterizes proof delay, allowing the proving stage to be studied without requiring full circuit execution in every benchmark run.

Submitter and Layer-1 Contract: The submitter handles bit-packing and compression strategies to commit data to Layer-1. It supports abstractions for different DA providers, routing data as calldata, blobs, or off-chain DA. The L1 smart contract acts as the verifier abstraction.

Telemetry: Granular data collection is embedded throughout to monitor queueing delays, prover utilization, and submission costs.

IV. EXPERIMENTAL METHODOLOGY

A. Study Objective and Scope

The objective is to systematically evaluate ZK-Rollup performance trade-offs using controlled configuration sweeps. The scope assumes a simplified operation set via synthetic token transfers, measuring relative performance shifts rather than absolute production metrics.

TABLE I
EXPERIMENTAL CONFIGURATIONS USED FOR PROTOTYPE EVALUATION

Config	Batch Policy	Seq. Policy	DA Mode	Workload	Prover Mode
C1	Fixed-size	FCFS	Calldata	Steady-state	Mock
C2	Fixed-size	Fee-priority	Calldata	Steady-state	Mock
C3	Soft-cap	Time-boost	Calldata	Bursty	Mock
C4	Soft-cap	FCFS	Blob	Bursty	Mock
C5	Tx-aware	Fair BFT	Blob	Skewed	Mock
C6	Fixed-interval	Fee-priority	Off-chain	Steady-state	Mock

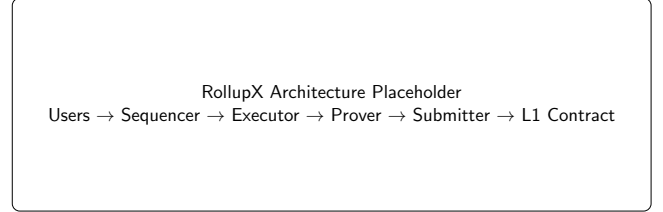


Fig. 2. Benchmark execution pipeline used for controlled configuration sweeps.

B. Experimental Factors and Regimes

The framework exposes configurable parameters (Table I), including batching policies (fixed-size, fixed-interval, soft-cap, transaction-aware), sequencing policies, and DA modes. Workload regimes cover balanced steady-state, heavy-transaction skew, and bursty arrivals with latency SLAs.

C. Cost Model and Metrics

The total transaction fee is modeled as the sum of the L2 fee (execution/proving resource usage) and the L1 fee (blob or calldata cost, commitment cost, verification cost).

The evaluation collects the following metrics:

- Submitted and Committed TPS
- Average and P95/P99 Latency
- Cost and Gas per Transaction
- Proof/Mock-Proof and Finalization Time
- DA Bytes and Queueing Delay
- Prover Utilization and Expired Transactions

D. Procedure and Hypotheses

The procedure involves initializing the prototype, applying the workload generator, and aggregating telemetry (Fig. 2). The experiments test hypotheses such as: agnostic batching degrades with heterogeneity, soft-cap improves robustness but may waste capacity, and transaction-aware batching dominates after a heterogeneity threshold.

V. RESULTS AND DISCUSSION

A. Preliminary Evaluation

The evaluation framework is prepared to report multi-dimensional metrics across the configured experimental space. Final benchmark values will be inserted after completing the full experiment suite. Table II illustrates the expected reporting format.

TABLE II
SUMMARY OF PRELIMINARY EXPERIMENTAL RESULTS

Config	Comm. TPS	Avg. Lat.	P95 Lat.	Cost/Tx	Proof Time	DA Bytes
C1	TBD	TBD	TBD	TBD	TBD	TBD
C2	TBD	TBD	TBD	TBD	TBD	TBD
C3	TBD	TBD	TBD	TBD	TBD	TBD
C4	TBD	TBD	TBD	TBD	TBD	TBD
C5	TBD	TBD	TBD	TBD	TBD	TBD
C6	TBD	TBD	TBD	TBD	TBD	TBD

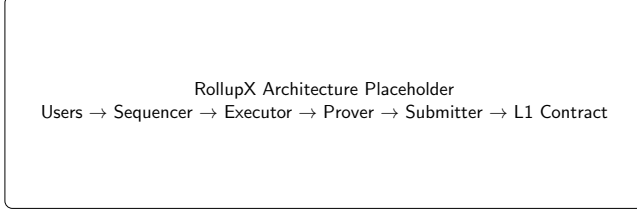


Fig. 3. Reporting template for throughput and latency comparison across configurations.

B. Throughput, Latency, and Cost Trade-offs

Preliminary analysis suggests that increasing batch size amortizes Layer-1 submission costs, decreasing the cost per transaction. However, this increases queuing delay, demonstrating a strict latency-cost trade-off. Furthermore, sequencing fairness (e.g., FCFS) often conflicts with revenue-optimizing policies (e.g., fee-priority), affecting transaction latency variance. Data availability choices further impact costs; blob-based DA is expected to drastically reduce the L1 footprint compared to standard calldata.

C. Design Implications

These interactions highlight several implications for rollup designers. TPS alone is insufficient to capture rollup performance; it must be contextualized with latency and cost.

- Batch sizes should be chosen based on the application’s latency tolerance.
- Transaction-aware batching is beneficial when workloads are heterogeneous.
- Prover utilization must be monitored as a first-class metric, as proof-generation can easily become the primary bottleneck.
- The choice of DA mode should align with specific security and cost assumptions.
- Soft finality (L2 confirmation) should be distinctly separated from L1 finality in reporting.

VI. CONCLUSION AND FUTURE WORK

This paper presented a modular ZK-Rollup research prototype designed to enable controlled experimental measurement of Layer-2 scalability bottlenecks. By isolating the sequencer, executor, prover, submitter, and data availability components, the framework supports reproducible configuration-level benchmarking. The defined methodology and preliminary evaluation highlight the complex trade-offs between throughput, latency, gas cost, and proof overhead.

A. Limitations

The proposed prototype is a research tool, not a production-ready system or a full zkEVM. It currently relies on synthetic workloads and a simplified operation set within a local testnet environment. Furthermore, the use of a mock prover simulates computational delay but does not capture the full cryptographic complexity of real circuit execution.

B. Future Work

Future work will focus on integrating full cryptographic prover capabilities and proof aggregation to measure exact hardware overheads. The framework will be extended to evaluate real trace-based workloads, support a complete trustless deposit and withdrawal path, and experiment with decentralized sequencers and stronger DA backends (e.g., Celestia or Avail). Ultimately, a public reproducibility package will be released to support the broader research community.

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